Reward-Free World-Model Pretraining for Transferable Control Representations

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Abstract

A world model trained without any task reward should, in principle, learn dynamics that transfer to many downstream tasks. Whether it actually does, and which property of the learned representation tracks that transfer, remains poorly characterized. We study reward-free pretraining of a DreamerV3 world model on the DeepMind Control proprioceptive Walker, comparing four pretraining strategies (none, random-policy data, Plan2Explore latent-disagreement, and APT particle-entropy) and adapting each to three downstream tasks (`stand`, `walk`, `run`) under a frozen-readout protocol. Across two seeds per condition we find that exploratory pretraining (Plan2Explore, APT) speeds early adaptation by roughly 2–3$\times$ over training from scratch, whereas random-policy pretraining gives no benefit and, when frozen, actively hurts. Adaptation speed correlates far more strongly with the coverage of the pretraining data and with linear probes of derived dynamical quantities (time-to-fall) than with current-state decodability, which is saturated and non-discriminative. A time-resolved analysis shows that Plan2Explore's coverage is built continuously by the learned exploratory policy, accumulating well after its raw disagreement reward has collapsed, which argues against a sustained, high-magnitude raw-disagreement reward as the driver. A minimal protocol-calibration experiment (Plan2Explore vs. no-pretraining on `walk`) shows the early-adaptation ordering carries over from frozen-readout to standard full-adaptation, while a frozen-readout-specific final-performance deficit disappears. The benefit is one of sample efficiency, not a higher performance ceiling. We frame our conclusions around what the small two-seed design can and cannot support, and are explicit that our coverage measure is nearly collinear with "the policy explores at all," so it cannot by itself establish coverage as the causal driver of transfer.

1 Introduction

Model-based reinforcement learning agents such as DreamerV3 (Hafner et al., 2023) learn a latent world model and train a policy entirely inside imagined rollouts. The world model is the agent's representation of its environment, and a natural question is whether that representation can be learned before any task is specified, then reused across tasks. This is the promise of unsupervised, or reward-free, reinforcement learning (Laskin et al., 2021): spend an exploration budget learning task-agnostic dynamics, and amortize it over many downstream objectives that share the same body and physics.

Two questions stand in the way of using this idea confidently. First, it is not obvious that reward-free pretraining helps at all in the model-based setting: a world model trained on a poor data distribution may model the wrong part of the state space, and a frozen representation may simply cap downstream performance. Second, even when pretraining helps, the field lacks a clear account of which property of the learned representation is responsible. Exploration methods are usually compared by their downstream return, not by the representations they induce, so it is rarely clear whether a method helps because it produces an accurate forward model, a linearly decodable state, or simply broad data coverage.

We take an empirical, mechanism-focused position on a deliberately small and controllable domain. Transfer here means reuse across reward functions on a single Walker body and proprioceptive observation space, not cross-morphology or cross-domain transfer. Because the 500K reward-free phase is a sunk cost paid once, any benefit is a per-task sample-efficiency gain that only amortizes across several downstream tasks, a point we make precise in Section 6. We pretrain a DreamerV3 world model without task reward under four strategies, adapt each to three Walker tasks, and pair the adaptation curves with a battery of representation probes and a direct measure of pretraining-data coverage. Our contributions are:

- A controlled four-condition comparison (no-pretrain, random, Plan2Explore, APT) on a shared body, isolating the effect of the pretraining exploration objective from the model and the downstream task.
- Evidence that exploratory pretraining improves early-adaptation sample efficiency by 2–3$\times$, that random pretraining does not, and that under a frozen representation random pretraining underperforms training from scratch.
- A representation analysis showing that adaptation speed tracks data coverage and derived-quantity probes, not current-state decodability, and a time-resolved coverage analysis that argues against a sustained raw-disagreement reward as the explanation for Plan2Explore's benefit.
- A protocol-calibration experiment separating "representation as fixed features" from "representation as initialization," which reconciles an apparent final-performance deficit of frozen pretraining.

Given the two-seed design and six-point correlations, these results should be read as descriptive evidence that motivates follow-up experiments rather than as significance claims.

2 Background and Related Work

World models and DreamerV3. The Dreamer family (Hafner et al., 2020; 2021; 2023) learns a Recurrent State-Space Model (RSSM) (Hafner et al., 2019) that encodes observations into a latent with a deterministic recurrent part and a stochastic categorical part, and trains an actor and critic on imagined latent rollouts. The world model is optimized for reconstruction and latent prediction, so it is in principle reusable across tasks that share dynamics; only the reward predictor and behavior are task-specific. We use DreamerV3 because its objective separates the task-agnostic world model from task-specific heads, matching the separation required by a frozen-readout transfer study.

Intrinsic-motivation exploration. When no task reward is available, exploration must be driven by an intrinsic signal. Two families are relevant here. Disagreement methods reward an agent for visiting states where an ensemble of forward predictors disagrees, a proxy for epistemic (reducible) uncertainty (Pathak et al., 2019); Plan2Explore (Sekar et al., 2020) brings this idea into the Dreamer setting by training the disagreement ensemble on latent features and optimizing the exploration policy in imagination. Entropy methods instead reward state-space coverage directly; APT (Liu & Abbeel, 2021b) maximizes a particle-based k-nearest-neighbor estimate of latent entropy (Singh et al., 2003). Related signals include prediction-error curiosity (Pathak et al., 2017) and random network distillation (Burda et al., 2019). A separate URLB family pursues behavioral diversity or successor features rather than novelty per se (e.g., DIAYN, Eysenbach et al., 2019; APS, Liu & Abbeel, 2021a); we restrict attention to Plan2Explore and APT because they instantiate the uncertainty-versus-coverage contrast within a single world-model agent with the architecture held fixed, which is what our representation comparison requires. Plan2Explore and APT are thus the natural contrast for our central question: does *uncertainty-driven* exploration induce a different, more transferable representation than *coverage-driven* exploration?

Unsupervised RL and adaptation protocols. The Unsupervised Reinforcement Learning Benchmark (Laskin et al., 2021) standardizes the two-stage recipe we follow: a long reward-free pretraining phase followed by a short reward-driven adaptation phase, on shared-body domains including Walker. URLB adapts by continuing to train the whole agent (*full-adaptation*). We instead make *frozen-readout* primary (the encoder and RSSM are frozen and only a fresh task head is trained) because it measures the representation as a fixed feature space rather than as an initialization, and then calibrate the two protocols against each other. A recurring URLB finding is that representation-centric methods can rank differently under the two protocols, which our calibration revisits directly.

Probing representations. Linear probes (Alain & Bengio, 2016) are a standard tool for asking what a frozen representation encodes. In a proprioceptive control setting the observation is already close to the simulator state, so probing the current state is near-trivial; we therefore emphasize future-state probes, derived dynamical quantities, and reward decodability, which are not linear read-outs of a single frame.

3 Research Questions and Approach

We address three questions: **(RQ1)** Does reward-free world-model pretraining improve downstream adaptation, and does the exploration objective matter? **(RQ2)** Which property of the pretrained representation is most associated with adaptation speed: forward-predictive accuracy, decodability of state and derived quantities, reward decodability, or data coverage? **(RQ3)** Do uncertainty-driven (Plan2Explore) and coverage-driven (APT) exploration induce measurably different transfer?

Conditions. All conditions share the DreamerV3 model and the Walker body, and differ only in how the world model is pretrained for 500K reward-free environment steps: **C1 (no-pretrain)** trains DreamerV3 from scratch on the downstream task; **C2 (random)** trains the world model on data from a uniform-random policy, with no actor; **C3 (Plan2Explore)** adds an ensemble of one-step latent predictors and trains an exploration actor on their disagreement; **C4 (APT)** trains an exploration actor on the k-NN particle-entropy of the latent. C1 is the no-pretraining baseline (not a strict lower bound, since at the adaptation budget it can beat a frozen pretrained model on final return); C2 isolates "a world model trained on passive data" from the effect of active exploration.

Intrinsic rewards (implementation). For C3 the ensemble predicts the next *deterministic posterior feature* (the recurrent state concatenated with the categorical posterior probabilities) rather than the sampled one-hot latent. We found the sampled-latent target degenerate: its irreducible sampling noise floors the ensemble loss while the optimal predictor collapses to the posterior mean, so disagreement vanishes even though the loss does not. Predicting the deterministic feature removes that aleatoric noise, leaving epistemic disagreement. Members use independent initialization and bootstrap masking. For C4 the reward at each imagined latent is $\log(c + \frac{1}{k}\sum_{j\in\mathrm{kNN}}\|z - z_j\|)$ computed over the imagination batch. Both intrinsic rewards are stop-gradient'd and consumed in place of the task reward head during imagination. Both are *adapted* variants of the originals: canonical Plan2Explore predicts the next encoder embedding and APT estimates entropy in a separate encoder space, whereas we predict the RSSM posterior feature and estimate entropy in the world-model latent. These choices fit the proprioceptive, latent-only setting, but cross-paper comparisons of absolute numbers should be read with the adaptation in mind.

Adaptation protocol. Frozen-readout is primary: we load the pretrained encoder, RSSM, and decoder, freeze them, and train a fresh actor, critic, and reward predictor on the downstream reward, using DreamerV3's imagination through the frozen RSSM. The adaptation budget is $A = 250\mathrm{K}$ steps. The **primary metric** is early-adaptation AUC: the mean episode return over the first 125K steps (the first half of A), equivalently a normalized area under the return curve, which reflects both how high adaptation starts and how fast it climbs. The **secondary metric** is final return near A. We report both because they answer different questions: how fast versus how high. With only two seeds per condition we report per-seed values and seed spread rather than significance tests, and make no significance claims, in keeping with small-sample reporting guidance for deep RL (Agarwal et al., 2021).

Representation analysis. We collect one common held-out trajectory set from a fixed uniform-random policy with logged MuJoCo physics, and encode it through each condition's frozen world model, so all conditions are probed on identical underlying states. On these features we fit ridge probes for current and future simulator state, gait phase, time-to-fall, and per-task reward, reporting held-out R^2. Separately, we measure coverage of each condition's pretraining replay as a k-NN

Table 1: Early-adaptation AUC (mean episode return over the first 125K steps; mean $\pm$ half the seed range, $n = 2$). Higher is faster adaptation.

Condition	`stand`	`walk`	`run`
C1 no-pretrain	321 ± 58	180 ± 59	52 ± 3
C2 random	312 ± 49	175 ± 6	63 ± 13
C3 Plan2Explore	676 ± 39	$\mathbf{381 \pm 39}$	141 ± 3
C4 APT	$\mathbf{740 \pm 36}$	323 ± 59	$\mathbf{167 \pm 19}$

particle entropy of the proprioceptive observations (an almost-complete function of the simulator state in this domain), on a shared standardizer so conditions are comparable. To probe mechanism we also bucket each replay buffer by collection time and recompute coverage per window.

4 Experimental Setup

Environment and model. We use the DeepMind Control Suite (Tassa et al., 2018) Walker in proprioceptive mode (no pixels; observation = orientations, torso height, velocities), simulated in MuJoCo (Todorov et al., 2012). The agent is the public DreamerV3 (JAX) at its smallest preset (`size1m`: RSSM deterministic state 512, 32 categorical latents of 4 classes, 64-unit MLPs), trained at `train_ratio` 1024. Pretraining is 500K reward-free steps; adaptation is 250K steps; C1 from-scratch runs to 500K, which also provides a full-performance reference.

Conditions, tasks, seeds. Four conditions $\times$ three tasks, two seeds each. C1/`walk` reuses two existing 500K from-scratch Walker-walk runs, so no new C1/`walk` job is needed. This gives 6 reward-free pretraining runs, 4 new C1 runs, and 18 frozen-readout adaptation runs. All metrics are reported with the per-seed spread, never as single point estimates.

Protocol calibration. To test whether frozen-readout conclusions transfer to the standard URLB protocol, we additionally run C3 (Plan2Explore) full-adaptation on `walk` for two seeds, in which the world model continues to train during adaptation. This is the minimal version of the calibration; a full four-way calibration is left to future work.

5 Results

5.1 RQ1: Exploratory pretraining speeds adaptation; random pretraining does not

Table 1 reports the primary metric. Exploratory pretraining (C3, C4) improves early-adaptation AUC over training from scratch by roughly 2–3$\times$ on every task: on `stand`, C1 reaches 321 versus 676 (C3) and 740 (C4); on `run`, 52 versus 141 and 167. Random pretraining (C2) is essentially identical to C1 ($\approx$ 312 vs. 321 on `stand`). We state these as multiplicative ranges over two-seed means and report the per-seed values in Appendix Table 5; with two seeds we claim a consistent direction and rough magnitude, not statistical significance. Figure 1 shows the same picture per task, and Figure 6 shows the underlying learning curves: the exploratory conditions start higher and climb faster.

The secondary metric tells a complementary and important story (Table 2). By 240K steps, from-scratch C1 has largely caught up: on `walk` it reaches 754, ahead of frozen C3 (535) and C4 (765, essentially tied). The pretraining benefit is therefore primarily one of sample efficiency rather than asymptotic performance at this budget. Two findings stand out. First, frozen C2 (random) collapses far below C1 on every task (144 vs. 754 on `walk`); its early and final values are nearly equal, meaning it plateaus almost immediately. A frozen world model trained on random-policy data, which mostly captures a flailing or fallen Walker, is a worse fixed representation than learning from scratch. Second, frozen C3 trails C1 at convergence on `walk` and `run` even though it adapts faster early, a tension we resolve in Section 5.4.

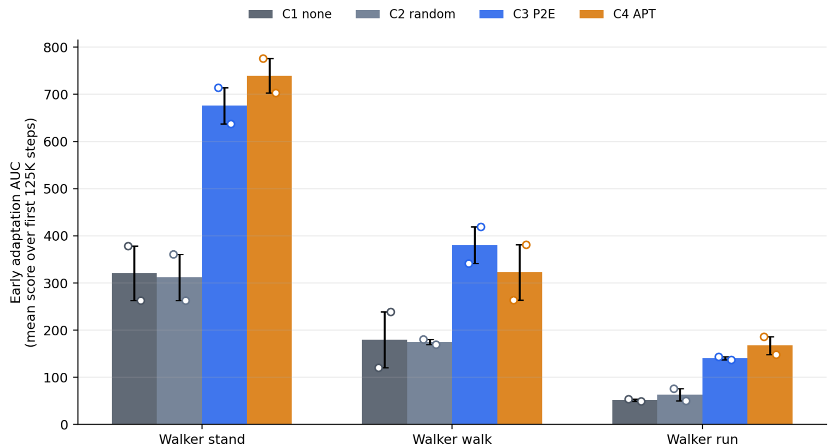

Figure 1: Early-adaptation AUC by condition and task. Plan2Explore and APT improve over both C1 (no-pretrain) and C2 (random); random pretraining matches the from-scratch baseline. Bars show the two-seed mean; markers show individual seeds.

Table 2: Final return (mean episode return over steps 220–240K; mean $\pm$ half the seed range). C1 is measured at the same step budget.

Condition	`stand`	`walk`	`run`
C1 no-pretrain	877 ± 36	754 ± 19	246 ± 54
C2 random	325 ± 48	144 ± 4	90 ± 35
C3 Plan2Explore	887 ± 69	535 ± 89	176 ± 31
C4 APT	929 ± 19	765 ± 78	243 ± 40

5.2 RQ2: Coverage and derived-quantity probes track transfer; current state does not

Table 3 summarizes the representation analysis on the common held-out set, and Figure 2 visualizes it. The current-state probe is saturated: every condition decodes the simulator state with $R^2 \approx 0.94$, and random pretraining is in fact marginally highest (0.953). Current linear decodability therefore carries essentially no signal about transfer in this proprioceptive domain. The strong negative correlation between the current-state probe and AUC ($r \approx -0.88$) is an artifact of this saturation (all values near a ceiling, the slowest condition marginally above it) and should not be read as a real effect.

The discriminative metrics are data coverage and the derived-quantity probes. Coverage (particle entropy of the pretraining replay) orders the conditions random $1.10 <$ APT $1.26 <$ Plan2Explore 1.39, and the time-to-fall probe R^2 orders them similarly ($0.55 < 0.63 \leq 0.65$). Across the six pretrained (condition, seed) points, early-adaptation AUC correlates $+0.68$ to $+0.90$ with coverage and $+0.67$ to $+0.90$ with time-to-fall (Table 3). Three caveats apply here. First, with $n = 6$ these correlations are descriptive, not significant; we read their sign and rough size, not their exact value. Second, and more fundamentally, our coverage measure is nearly collinear with "the policy explores at all": random barely moves and the two exploratory conditions both move, so any metric separating active from passive policies will correlate with AUC. With three conditions the correlation cannot isolate coverage as the *causal* driver; it largely restates the exploratory-versus-passive split the conditions were built to create. Third, the k=20 future-state probe appears with a high positive correlation in Table 3, but its absolute R^2 is within noise of zero for every condition, so that correlation rides on the sign of negligible numbers and is no more interpretable than the saturated current-state probe.

Table 3: Representation metrics on the common held-out set (condition means over 2 seeds), and their correlation with early-adaptation AUC across the six pretrained (condition, seed) points (range over the three tasks, $n = 6$). C1 has no pretrained world model and is omitted.

Metric	C2 random	C3 P2E	C4 APT	corr. with AUC
Current-state probe R^2	0.953	0.941	0.944	-0.85 to -0.90 (artifact)
Future-state probe R^2 (k=20)	-0.003	0.002	0.001	$+0.84$ to $+0.92$
Time-to-fall probe R^2	0.554	0.627	0.647	$+0.67$ to $+0.90$
Reward probe R^2 (`walk`)	0.615	0.651	0.641	$+0.48$ to $+0.62$
Coverage (particle entropy)	1.100	**1.386**	1.262	$+0.68$ to $+0.86$

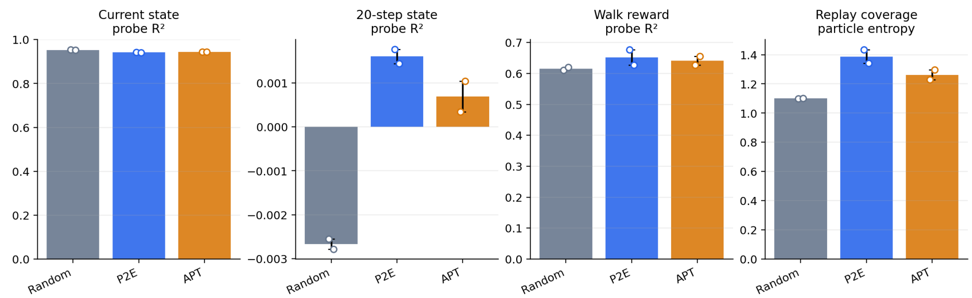

Figure 2: Representation summary. The current-state probe is saturated across conditions (left), while data coverage clearly separates the methods (right). Adaptation speed tracks the latter, not the former.

5.3 RQ3: Plan2Explore and APT transfer comparably, and coverage outlasts the disagreement reward

On adaptation, C3 and C4 are broadly comparable: APT leads on `stand` and `run`, Plan2Explore on `walk`, and their seed ranges overlap on `stand` and `walk`. We therefore cannot separate the two methods' downstream effect at two seeds, which is itself the answer to RQ3 at this budget: uncertainty-driven and coverage-driven exploration produce similar transfer here. The uncertainty-driven method (Plan2Explore) in fact produced the highest data coverage, more than the coverage-driven method (APT), so coverage and the nominal objective do not line up in the naive way.

The timing analysis sharpens the mechanism. Plan2Explore's raw disagreement reward collapses by roughly $100\times$ early in pretraining (from ~ 0.046 to ~ 0.0005 by 250K steps) as the world model reduces its uncertainty. Yet its replay coverage keeps rising monotonically through the end of pretraining (Figure 3, Figure 4), long after the reward has vanished, whereas random-policy coverage stays flat throughout. The *learned exploratory policy* thus generates this coverage continuously, well after the intrinsic reward has decayed, rather than in an early burst that is later retained. This argues against a sustained, high-magnitude *raw*-disagreement reward as the driver, though we measured the raw signal, not the percentile-normalized advantage the actor actually optimizes, so a normalized or relative-uncertainty effect is not excluded. The coverage account is consistent with the data, but does not prove it, and we give equal weight to a competing explanation: the exploratory policy becomes an active, upright mover, so its world model simply sees the task-relevant locomotion regime that the random policy never reaches. Under this view the operative variable is not breadth of coverage per se but whether pretraining reaches the right region of state space at all. Our data cannot separate "broad coverage" from "reached the task-relevant regime," because the conditions that have one have the other; distinguishing them (Section 6) is the natural next step.

5.4 Protocol calibration: the early-adaptation gap survives full-adaptation; the final-return deficit does not

Table 4 compares C1 and C3 on `walk` under both protocols. On the primary metric the two protocols agree for this C1-vs-C3 contrast: C3 beats C1 early under frozen-readout (+200 AUC) and by even

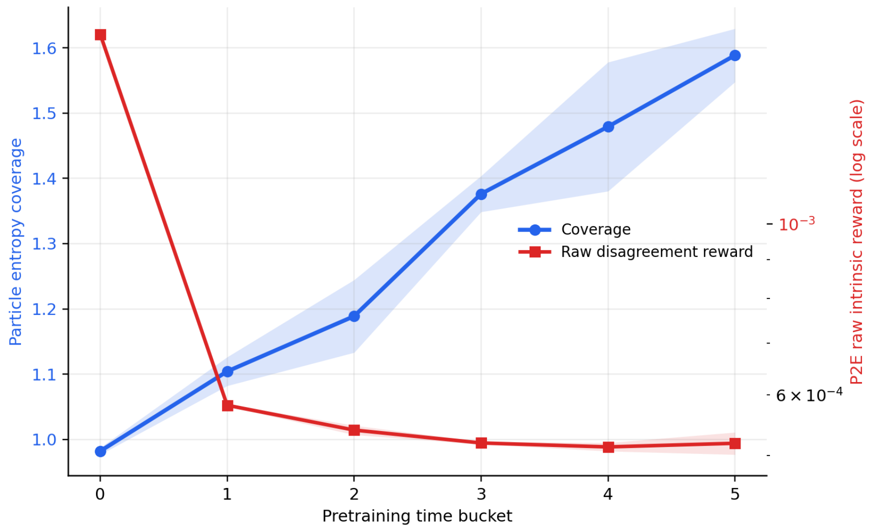

Figure 3: Plan2Explore timing. The raw disagreement reward collapses early in pretraining, while replay coverage continues to grow, indicating the useful data distribution is sustained by the learned policy rather than by the reward signal.

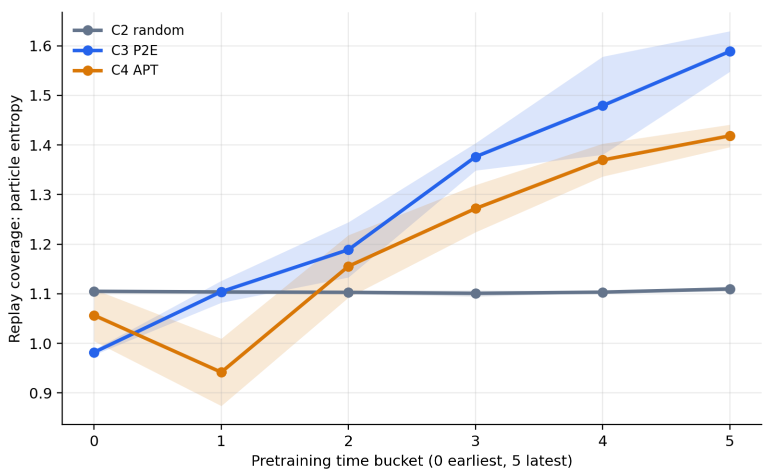

Figure 4: Coverage over pretraining time, bucketed by collection step. Random coverage is flat; Plan2Explore and APT coverage rise monotonically and peak in the final window.

more under full-adaptation (+382), so the early-adaptation gap is not a frozen-readout artifact for the one contrast we calibrated. We do not claim protocol-invariance beyond C1-vs-C3 on `walk`. On final return the ordering flips: under freezing C3 trails C1 by 219, but under full-adaptation C3 leads by 191 and reaches near-ceiling (945). The frozen-readout final deficit reflects the cost of holding the world model fixed, not a deficiency of the pretrained dynamics. We did not run C2 full-adaptation, so we do not extend this explanation to C2's collapse, which under freezing could equally reflect genuinely poor random-policy dynamics. Plan2Explore's world model is a strong *initialization* even where it is a mediocre *fixed feature space*, precisely the "representation as initialization vs. fixed features" distinction the calibration was designed to expose. Figure 5 visualizes the three curves.

6 Discussion and Conclusion

We asked whether reward-free world-model pretraining yields transferable control representations, which representational property tracks the transfer, and whether the exploration objective matters.

Table 4: Protocol calibration on `walk` (2 seeds). For the C1-vs-C3 contrast, the early-AUC ordering matches across protocols; the final-return ordering does not.

Setting	Early-AUC	Final return
C1 from-scratch (inherently full)	180	754
C3 frozen-readout	381	535
C3 full-adaptation	**562**	**945**
C1→C3 gap (frozen)	+200	−219
C1→C3 gap (full)	+382	+191

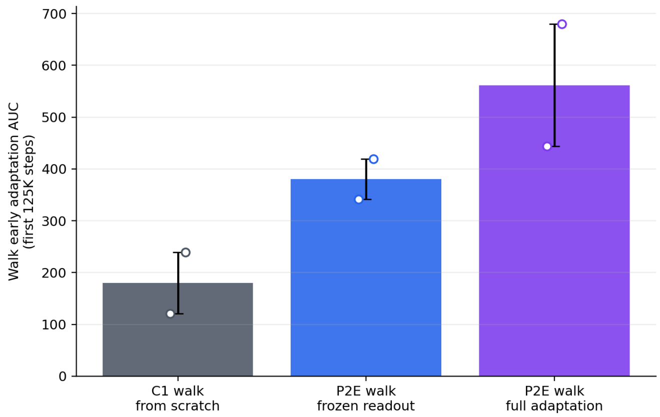

Figure 5: Walk adaptation under both protocols. Full-adaptation C3 dominates both frozen C3 and from-scratch C1 on speed and final return.

On a controlled DMC Walker study with four pretraining conditions, three tasks, and matched probes, we find that exploratory pretraining (Plan2Explore, APT) improves early-adaptation sample efficiency by 2–3×, that random-policy pretraining does not and is actively harmful as a frozen representation, and that adaptation speed tracks the coverage of the pretraining data and derived-quantity probes rather than the saturated current-state probe (RQ1, RQ2). Uncertainty-driven and coverage-driven exploration transfer comparably (RQ3), and a time-resolved analysis argues against a sustained, high-magnitude raw-disagreement reward as the driver for Plan2Explore: its coverage is built continuously by the learned policy, even after its raw disagreement reward collapses. A minimal calibration (C1 vs. C3 on `walk`) shows that the early-adaptation ordering carries over to standard full-adaptation, while a frozen-readout-specific final-performance deficit disappears.

Implications. The results suggest a few lessons beyond this Walker setup. First, in proprioceptive control the representation property that tracks transfer is not the linear decodability of the current state, which saturates, but the data distribution the exploration policy induces (which here we cannot cleanly separate from simply reaching the task-relevant regime). Either way, this argues for evaluating exploration methods by the distribution they collect, rather than by downstream return alone. Second, the choice of adaptation protocol is itself a finding, not a nuisance: frozen-readout is a conservative stress test that can invert final-performance orderings relative to full-adaptation, so transfer claims should state which protocol they assume. Third, because the gain is sample efficiency rather than a higher ceiling, a 500K reward-free pretraining phase only "pays for itself" when amortized across several downstream tasks: it is a one-time cost that buys a faster start on each task, and at our budgets the early saving is on the order of 10^5 steps per task, so the economics favor pretraining only when the number of downstream tasks sharing the body is large enough to outweigh the pretraining steps.

Limitations. The study uses two seeds per condition and six points per correlation, so all correlations are descriptive rather than significant and the C3-vs-C4 ordering is unresolved; the seed spread we report makes this visible. The coverage-correlation is partly confounded by the binary exploratory-vs-passive split, and because our coverage measure is nearly collinear with "the policy is an active explorer," the analysis cannot attribute transfer to coverage as opposed to simply reaching the task-relevant regime. Coverage-mediation of transfer therefore remains a hypothesis: the clean causal test (adapting from intermediate pretraining checkpoints) requires checkpoint retention we did not enable and is left to future work. The protocol calibration is minimal (Plan2Explore on `walk` only); a full four-way, three-task calibration would be needed to claim protocol-invariance broadly. The pretraining budget (500K) is short relative to URLB's $\sim$2M; this especially weakens the negative C2 result, since a random policy might collect more useful data given much longer, so we read "random pretraining does not help" as budget-conditional rather than absolute. Finally, the domain is a single shared body, and with two seeds we report direction and spread rather than significance, so absolute magnitudes and the unresolved C3-versus-C4 ordering should not be over-interpreted.

Future work. The most informative next experiment is adaptation from time-stamped pretraining checkpoints, which would convert the coverage-timing observation into a causal test. Extending the four-way calibration and adding seeds would resolve C3 versus C4. Finally, disentangling "broad coverage" from "task-relevant active locomotion" (for example by measuring coverage restricted to upright, moving states) would clarify whether exploration helps by breadth or by reaching the right regime.

Reproducibility

Code for world-model pretraining, adaptation, probing, coverage, and synthesis is available at `https://github.com/shalalababa/dreamerv3/tree/arl-exploration`. The branch contains the implementation and report figures used here; raw run outputs are omitted from the repository. Reported numbers are computed by the probing pipeline (`probing/`) from logged run metrics; tables cite condition means with per-seed spread.

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A PER-SEED RESULTS

Table 5 gives the per-seed early-adaptation AUC behind the condition means in Table 1, showing the two-seed spread directly.

Table 5: Per-seed early-adaptation AUC.

Condition	Seed	`stand`	`walk`	`run`
C1	1	378.8	120.9	54.4
C1	2	263.3	239.2	49.1
C2 random	1	361.3	181.0	76.5
C2 random	2	262.8	169.7	50.0
C3 P2E	1	714.7	341.6	144.2
C3 P2E	2	637.1	419.5	137.2
C4 APT	1	775.8	264.2	185.9
C4 APT	2	703.5	381.4	148.7

Table 6: Per-seed final return (mean episode return over steps 220–240K), behind the condition means in Table 2.

Condition	Seed	`stand`	`walk`	`run`
C1	1	913	735	300
C1	2	841	773	192
C2 random	1	372	148	125
C2 random	2	277	140	55
C3 P2E	1	955	624	207
C3 P2E	2	818	446	145
C4 APT	1	948	687	282
C4 APT	2	910	844	203

Table 7: Per-seed representation metrics on the common held-out set (probe R^2 and replay coverage), behind the condition means in Table 3. C1 has no pretrained world model. The future-state probe (k=20) is omitted because its R^2 is within noise of zero for every condition and seed (cf. Table 3) and is therefore not interpretable per seed.

Cond.	Seed	state $R^2_{k=0}$	TTF R^2	rew$_{\text{walk}}$ R^2	cov.
C2 random	1	0.953	0.578	0.611	1.099
C2 random	2	0.953	0.530	0.620	1.102
C3 P2E	1	0.942	0.658	0.676	1.433
C3 P2E	2	0.941	0.597	0.626	1.340
C4 APT	1	0.944	0.632	0.627	1.227
C4 APT	2	0.945	0.663	0.656	1.296

Full-adaptation calibration, per seed. C3 (Plan2Explore) full-adaptation on `walk`: seed 1 early-AUC 444, final 943; seed 2 early-AUC 679, final 948 (means 562 / 945 in Table 4).

B Hyperparameters and reproducibility details

- **Environment.** DMC Walker, proprioceptive observations (orientations, torso height, velocities; 24-d), action repeat 1, episode length 1000, MuJoCo physics.
- **Agent.** DreamerV3 `size1m`: RSSM deterministic state 512, 32 stochastic categoricals of 4 classes, encoder/decoder MLPs 64 units $\times$ 3 layers, symlog inputs. Batch 16×64; optimizer learning rate 4×10^{-5}; `train_ratio` 1024; 16 parallel envs; imagination horizon 15; discount horizon 333.
- **Budgets / seeds.** Pretraining 500K reward-free steps; adaptation $A = 250$K; C1 from-scratch to 500K. Two seeds per condition, RNG seeds $\{1, 2\}$.
- **Plan2Explore (C3).** Ensemble of 8 one-step predictors (2 layers $\times$ 256 units), target = next deterministic posterior feature (deter $\oplus$ softmax-posterior, 640-d), bootstrap-mask probability 0.8, intrinsic-reward scale 10^3 before percentile normalization.
- **APT (C4).** k-NN particle entropy over the 640-d world-model latent with $k = 12$, $c = 1.0$, computed within each imagination batch.
- **Frozen-readout.** Encoder + RSSM + decoder loaded from the pretraining checkpoint (parameter regex `^(enc|dyn|dec)/`) and frozen; a fresh actor, critic, and reward head are trained.
- **Probes.** Common held-out set: 40 episodes from a uniform-random policy with logged physics. Ridge probes with $\lambda \in \{10^{-3}, 10^{-1}, 1, 10, 10^3\}$ selected on a 20% validation split; test fraction 0.3; sites encoder / posterior / prior / open-loop imagination at $k \in \{1, 5, 20\}$; target horizons $\{0, 1, 5, 20\}$.
- **Coverage.** Particle entropy with $k = 12$, $c = 1.0$, up to 3000 frames per buffer; 2-D histogram entropy over the top two principal components (24 bins); a single standardizer and PCA fit on the pooled frames so conditions share a scale; 6 equal time buckets for the coverage-over-time curve.
- **Metrics.** Episode return is logged at each episode completion. Early-adaptation AUC is the trapezoidal integral of (step, episode-return) over the first 125K steps divided by the integrated step span (a normalized mean); final return is the mean episode return over steps 220–240K. Both are computed per seed and reported as the two-seed mean with the seed half-range.

C Adaptation learning curves

Figure 6 shows the full adaptation learning curves underlying the AUC summary.

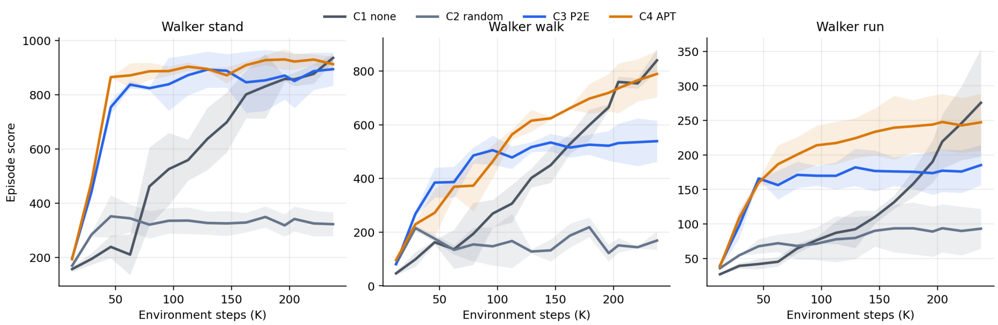

Figure 6: Adaptation learning curves (return vs. environment steps) by condition and task. Exploratory conditions start higher and rise faster; from-scratch C1 converges high by the end of the budget; frozen random plateaus low.